

Math 74: Algebraic Topology

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Problem 1. Show that for a space X , the following three conditions are equivalent:

1. Every map $S^1 \rightarrow X$ is homotopic to a constant map, with image a point.
2. Every map $S^1 \rightarrow X$ extends to a map $D^2 \rightarrow X$.
3. $\pi_1(X, x_0) = 0$ for all $x_0 \in X$.

Deduce that a space X is simply-connected iff all maps $S^1 \rightarrow X$ are homotopic. [In this problem, 'homotopic' means 'homotopic without regard to basepoints'.]

Solution.

$$\begin{array}{ccc}
 S^1 \times [0, 1] & \xrightarrow{H} & X \\
 \pi \downarrow & \nearrow f' & \\
 D^2 & & \\
 \iota \uparrow & \nwarrow f & \\
 S^1 & &
 \end{array}$$

(1 $\implies$ 2)

Assume every map $f : S^1 \rightarrow X$ is homotopic to a constant map e_{x_0} . Let $H : S^1 \times [0, 1] \rightarrow X$ be the homotopy between f and e_{x_0} , with $H(s, 0) = f(s)$ and $H(s, 1) = e_{x_0}(s)$. We define a quotient map,

$$\pi : S^1 \times [0, 1] \rightarrow D^2, \quad \pi(s, t) = (1 - t)s$$

Note that π is continuous, surjective and maps $S^1 \times \{1\}$ to a constant point. Since H is constant on $S^1 \times \{1\}$, it induces (via π) a continuous map $f' : D^2 \rightarrow X$. Note that,

$$f' \circ \pi(s, 0) = f'((1 - 0)s) = f'(s) = H(s, 0) = f(s)$$

That is $f'(s) = f(s)$ on $s \in \partial D^2 = S^1$. Hence, every map f extends to a map f' from D^2 to X .

(2 $\implies$ 3)

Assume that every map $f : S^1 \rightarrow X$ extends to a map $f' : D^2 \rightarrow X$.

Let $\iota : S^1 \rightarrow D^2$ be the natural inclusion map. Then, $f = f' \circ \iota$, and $f_* = f'_* \circ \iota_*$. But,

$$\iota_* : \pi_1(S^1, s_0) \rightarrow \pi_1(D^2, d_0)$$

is trivial because $\pi_1(D^2, d_0) = \{0\}$. Therefore $\iota_*([\beta]) = 0$ for all $[\beta] \in \pi_1(S^1, s_0)$. Then,

$$f_*([\beta]) = f'_* \circ \iota_*([\beta]) = f'_*(0) = 0$$

Now, note that every loop $[\gamma] \in \pi_1(X, x_0)$ can be represented by some map $\gamma : S^1 \rightarrow X$, and by assumption it extends to some $f' : D^2 \rightarrow X$. Applying the argument above gives us $[\gamma] = 0$ and hence $\pi_1(X, x_0) = 0$ for all $x_0 \in X$.

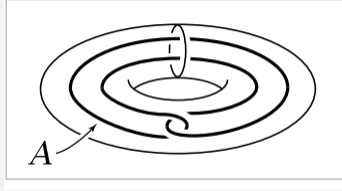
(3 $\implies$ 1)

Assume $\pi_1(X, x_0) = 0$ for all $x_0 \in X$.

Let $s_0 \in S^1$ be the basepoint and pick $x_0 = f(s_0)$. Since f is a map from $S^1 \rightarrow X$ it is a loop and since that every loop $[\gamma] \in \pi_1(X, x_0)$ can be represented by some map $\gamma : S^1 \rightarrow X$ we know it is homotopic to the constant loop $[e_{x_0}] \in \pi_1(X, x_0)$. And hence every map $f : S^1 \rightarrow X$ is homotopic to a constant map e_{x_0} .

Thus the three statements are equivalent.

Problem 2. Show that there are no retractions $r : X \rightarrow A$ where $X = S^1 \times D^2$ and A the circle shown in the figure.



Solution.

Problem 3. Show that the complement of a finite set of points in $\mathbb{R}^n$ is simply-connected if $n \geq 3$.

Solution.

Inductive Case: Let P be the hyperplane separating x_k from $\{x_1, \dots, x_{k-1}\}$. Now let U be the open half-space containing x_k minus x_k , and V the open half-space containing $\{x_1, \dots, x_{k-1}\}$ minus $\{x_1, \dots, x_{k-1}\}$, chosen so that they overlap in a strip around the hyperplane P . That is, $U \cap V = P \times (-\epsilon, \epsilon)$ for some small $\epsilon > 0$. Then,

$$U \cup V = \mathbb{R}^n \setminus X \text{ and } U \cap V = P \times (-\epsilon, \epsilon) \cong \mathbb{R}^{n-1}$$

Then we have,

$$U \simeq \mathbb{R}^n \setminus \{x_k\} \simeq S^{n-1} \implies \pi_1(U) = \{0\}$$

(where $\pi_1(S^{n-1}) = 0$ since $n \geq 3$), and by the Inductive Hypothesis, since V is homotopy equivalent to $\mathbb{R}^n$ with $k-1$ points removed we have $\pi_1(V) = \{0\}$. Moreover, $\pi_1(P) = \{0\}$ since $P \cong \mathbb{R}^{n-1}$ and $\mathbb{R}^{n-1}$ is simply connected.

Then by van Kampen, we have,

$$\pi_1(\mathbb{R}^n \setminus X) \cong (\pi_1(U) \star \pi_1(V)) / N = \{0\}$$

where N is trivial since $\pi_1(U \cap V) = \{0\}$. Thus the complement of a finite set of points in $\mathbb{R}^n$ is simply connected if $n \geq 3$.

Solution.

Let $X = \{x_1, \dots, x_k\} \subset \mathbb{R}^n$. We prove $\pi_1(\mathbb{R}^n \setminus X) = \{0\}$ by induction on k .

Base Case: $k = 1$.

We know that $\mathbb{R}^n \setminus \{x\} \simeq S^{n-1}$ by deformation retraction, and that $\pi_1(S^{n-1}) = \{0\}$ since $n-1 \geq 2$. Hence, $\mathbb{R}^n \setminus \{x\}$ is simply connected.

Inductive Hypothesis. Assume that for any set of $k - 1$ points, $\pi_1(\mathbb{R}^n \setminus X) = \{0\}$.

Inductive Step. Since the x_i are distinct, choose a hyperplane $P \cong \mathbb{R}^{n-1}$ separating x_k from $\{x_1, \dots, x_{k-1}\}$. Define

$$U = (H^+ \cup (P \times (-\varepsilon, \varepsilon))) \setminus \{x_k\}, \quad V = (H^- \cup (P \times (-\varepsilon, \varepsilon))) \setminus \{x_1, \dots, x_{k-1}\},$$

where H^+ is the open half-space on the side of x_k , H^- is the open half-space on the side of $\{x_1, \dots, x_{k-1}\}$, and $\varepsilon > 0$ is picked to be small enough that the strip $P \times (-\varepsilon, \varepsilon)$ contains none of the x_i . Then U and V are open sets satisfying

$$U \cup V = \mathbb{R}^n \setminus X, \quad U \cap V = P \times (-\varepsilon, \varepsilon) \cong \mathbb{R}^{n-1}.$$

Note that since $n \geq 2$, removing finitely many points from an open subset of $\mathbb{R}^n$ preserves path-connectivity, so U , V , and $U \cap V \cong \mathbb{R}^{n-1}$ are all path-connected. Fix a basepoint $x_0 \in U \cap V$. We now compute the fundamental groups of each piece:

$$U \simeq \mathbb{R}^n \setminus \{x_k\} \simeq S^{n-1} \implies \pi_1(U) = \{0\},$$

where $\pi_1(S^{n-1}) = \{0\}$ since $n \geq 3$. By the inductive hypothesis, $V \simeq \mathbb{R}^n \setminus \{x_1, \dots, x_{k-1}\}$ gives $\pi_1(V) = \{0\}$. Moreover, $U \cap V \cong \mathbb{R}^{n-1}$ is simply connected, so $\pi_1(U \cap V) = \{0\}$.

Then by the Seifert–van Kampen theorem applied to the cover $\{U, V\}$,

$$\pi_1(\mathbb{R}^n \setminus X) \cong (\pi_1(U) \star \pi_1(V)) / N,$$

Since $\pi_1(U \cap V) = \{0\}$, the subgroup N is trivial, and so

$$\pi_1(\mathbb{R}^n \setminus X) \cong \{0\} \star \{0\} = \{0\}.$$

Thus the complement of any finite set of points in $\mathbb{R}^n$ is simply connected for $n \geq 3$.

Problem 4. Let $X \subset \mathbb{R}^3$ be the union of n lines through the origin. Compute $\pi_1(\mathbb{R}^3 - X)$.

Solution.

Problem 5. Proof for Theorem 1.38